

Fully Symmetric No-Three-in-Line Configurations: a Structural Reduction, Exhaustive Verification to $n = 64$, and the Failure of the First-Moment Heuristic

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Abstract

The no-three-in-line problem asks whether the $n \times n$ grid contains $2n$ points with no three collinear. We study the subclass of configurations invariant under the full dihedral symmetry group D_4 of the square, for which the known solutions occur at $n = 2, 4, 10$ and Guy and Kelly conjectured in 1968 that no others exist. We reduce the problem exactly to a search over involutions of $\{1, \dots, m\}$, where $n = 2m$, prove two arithmetic necessary conditions on such involutions — a Sidon-type condition on a set of m labels, which by itself rules out $n = 8$ by hand and isolates the $n = 10$ solution uniquely, and a distinctness condition on associated reduced fractions — and use an optimized backtracking search to verify independently that no further solutions exist for any even $n \leq 64$. We then show that the natural Guy–Kelly first-moment heuristic, adapted to this symmetry class, *over*-predicts solutions by an unboundedly growing factor. Exact enumeration of all involutions satisfying the Sidon condition through $m = 20$ reveals why: the minimum number of violated lines over the entire class follows a growing staircase $8, 16, 24, 32$, so the obstruction is extremal rather than probabilistic. We propose a certified-UNSAT SAT-solving campaign, sized at roughly 1,540 variables and 10^6 – 10^7 clauses at $n \approx 110$, as the practical route to extending the verified range beyond the reported frontier of $n = 108$, and we formulate a defect-floor conjecture whose proof would settle the original question.

1 Introduction

Dudeney’s 1906 puzzle asks for $2n$ points on the $n \times n$ grid with no three collinear; since n horizontal lines cover the grid and each may carry at most two chosen points, $2n$ is the trivial upper bound. Whether it is attained for every n is the no-three-in-line problem. Guy and Kelly [1] conjectured on probabilistic grounds that only finitely many n admit solutions, with at most cn points selectable asymptotically for a constant $c < 2$; their constant, as corrected by Ellmann (2004), is $\pi/\sqrt{3} \approx 1.814$. Solutions are nevertheless known for every $n \leq 70$ and for several larger sizes, the current record being $n = 74$ (Prellberg, July 2026; see [4]). The complete solution counts for $n \leq 20$ are in OEIS A000769. Activity in 2025–2026 has been intense: Prellberg and Riley enumerated the rotational classes `rot4` and `rct4` through the mid-50s using constraint programming and CUDA, and Heule applied SAT solvers to the same classes beginning June 2026 [4].

This report concerns the most symmetric class: configurations invariant under the full symmetry group of the square (Flammenkamp’s class `full`). Three such configurations are known, at $n = 2, 4$, and 10 , and:

Conjecture 1 (Guy–Kelly Conjecture I, 1968). *There are exactly three no-three-in-line configurations having the full symmetry of the grid.*

Flammenkamp’s near-miss computations (2006) found the maximal point count achievable with full symmetry drifting steadily away from $2n$ for even $n \leq 76$, and the class has reportedly been searched empty well beyond that (to $n = 108$; we did not independently verify the endpoint and treat it as reported). Our contributions: an exact structural reduction (§2); two arithmetic obstructions with short proofs (§3); independent exhaustive verification through $n = 64$ (§4); a quantitative analysis showing the first-moment heuristic fails here and why (§5); and concrete methods for extending the search beyond $n = 108$ (§6).

2 The involution model

Throughout, a *solution* means $2n$ grid points with no three collinear and full D_4 symmetry. Any solution has exactly two points in every row and every column.

Theorem 1. *No solution exists for odd n .*

Proof. Center the grid so that the reflections are $x \mapsto -x$ and $y \mapsto -y$. For odd n the center column $x = 0$ consists of lattice points. If $(0, y)$ is chosen, any second point (x, y) of its row forces its mirror $(-x, y)$, giving three in the row; so no point lies on the center column, contradicting the requirement of exactly two points per column. $\square$

For even $n = 2m$ we center the grid so that lattice points have both coordinates odd, in $\{\pm 1, \pm 3, \dots, \pm(n-1)\}$; we call these *doubled-odd coordinates*. D_4 acts by coordinate sign changes and the swap $(x, y) \mapsto (y, x)$.

Theorem 2. *Solutions with $n = 2m$ correspond bijectively to involutions σ of $\{1, \dots, m\}$ with exactly $m \bmod 2$ fixed points such that the $4m$ points*

$$P(\sigma) = \{ (\pm(2\sigma(i) - 1), \pm(2i - 1)) : i = 1, \dots, m \}$$

contain no three collinear points.

Proof. Each row $y = 2i - 1$ holds exactly two points, which the mirror $x \mapsto -x$ exchanges (a mirror-fixed point would need $x = 0$, not a lattice value); write them $(\pm x_i, 2i - 1)$ with x_i odd and positive. The mirror $y \mapsto -y$ forces rows $\pm(2i - 1)$ to share the same x_i . The diagonal reflection maps the point set to itself, which states precisely that $i \mapsto (x_i + 1)/2$ is an involution σ . A fixed point of σ contributes the four points $(\pm a, \pm a)$ with $a = 2i - 1$, of which two lie on each long diagonal; two fixed points would put four points on the main diagonal, which contains a collinear triple, so σ has at most one fixed point, and the parity of m forces exactly $m \bmod 2$. Conversely, any such σ with $P(\sigma)$ triple-free is a solution: rows and columns hold exactly two points by construction. $\square$

Corollary 1. *If $n \equiv 0 \pmod{4}$ the long diagonals are empty; if $n \equiv 2 \pmod{4}$ each long diagonal carries exactly two points.*

This matches Flammenkamp’s symmetry remarks [5], derived there differently. The reduction is what makes the class tractable: the configuration space at $n = 2m$ has size $(m-1)!!$ for even m and $m(m-2)!!$ for odd m — about e^{81} at $n = 108$ — so all progress must come from pruning, and the theorems identify exactly what object is being pruned.

3 Arithmetic obstructions

The two lemmas below translate specific line families into conditions on σ alone. Both were verified to be exactly necessary against full enumeration for $m \leq 13$.

Lemma 1 (Slope ± 1 ; the Sidon condition). *For a pair orbit $\{i, j\}$, $i < j$, the eight points of $P(\sigma)$ arising from i and j place exactly two points on each of the four lines $x - y = \pm 2(j - i)$ and $x - y = \pm 2(i + j - 1)$; a fixed orbit $\{f\}$ places one point on each of $x - y = \pm 2(2f - 1)$ and two on $x - y = 0$. Consequently, in any solution the m labels*

$$\mathcal{L}(\sigma) = \{j - i : \text{pairs}\} \cup \{i + j - 1 : \text{pairs}\} \cup \{2f - 1 : \text{fixed}\}$$

are pairwise distinct. Conversely, every violation on a line of slope ± 1 is a label collision, so the condition characterizes this family exactly.

Proof. Direct computation: the points $(2j - 1, 2i - 1)$ and $(-(2i - 1), -(2j - 1))$ lie on $x - y = 2(j - i)$, while $(2j - 1, -(2i - 1))$ and $(2i - 1, -(2j - 1))$ lie on $x - y = 2(i + j - 1)$; sign changes give the mirrored lines, and the diagonal reflection carries the slope $+1$ family to the slope -1 family. Since each orbit contributes at most two points to any such line, three points on one line require contributions from two orbits, i.e. a repeated label. Within one orbit $j - i = i + j - 1$ would force $i = \frac{1}{2}$. $\square$

The labels lie in $\{1, \dots, 2m - 1\}$, so a solution must realize m distinct values from a range of size $2m - 1$: a Sidon-flavored condition of density about one half, reminiscent of graceful labelings. It is remarkably restrictive in practice (Table 1), and at $m = 4$ it is already vacuous:

Corollary 2. *There is no fully symmetric solution for $n = 8$.*

Proof. $m = 4$ is even, so σ is one of the three pairings of $\{1, 2, 3, 4\}$. $(12)(34)$ has labels $\{1, 2\} \cup \{1, 6\}$; $(13)(24)$ has $\{2, 3\} \cup \{2, 5\}$; $(14)(23)$ has $\{3, 4\} \cup \{1, 4\}$. Each repeats a label. $\square$

At $m = 5$ exactly one involution satisfies Lemma 1, namely $\sigma = (1\ 5)(3\ 4)$ with fixed point 2 — and it is the classical 10×10 solution, with doubled-odd points $(\pm 9, \pm 1), (\pm 1, \pm 9), (\pm 7, \pm 5), (\pm 5, \pm 7), (\pm 3, \pm 3)$. The dominant constraint of the problem therefore has a purely arithmetic description.

Lemma 2 (Central lines). *Each pair orbit $\{i, j\}$ places two points on each line through the origin of slope $\pm(2i - 1)/(2j - 1)$ and $\pm(2j - 1)/(2i - 1)$, and a fixed orbit uses slopes ± 1 . In any solution the reduced fractions $(2i - 1)/(2j - 1)$, taken together with their reciprocals, are pairwise distinct across orbits.*

For example the orbits $\{2, 5\}$ and $\{1, 2\}$ are incompatible: $3/9$ and $1/3$ reduce to the same fraction.

4 Exhaustive verification through $n = 64$

We implemented a depth-first search over involutions in C. State is the set of placed points together with a hash table counting, for every line spanned by two placed points, the number of point pairs on it; a count reaching three signals a collinear triple. The DFS discipline permits strict LIFO undo: hash entries are chained with head insertion, so an entry to be deleted on backtrack is always its bucket's head, and both insertion and deletion are $O(1)$ with a stack

Table 1: Exact filter strength of Lemmas 1 and 2, and the true solution counts, by full enumeration.

n	m	involutions	pass S	pass R	pass S^R	solutions
2	1	1	1	1	1	1
4	2	1	1	1	1	1
6	3	3	1	3	1	0
8	4	3	0	3	0	0
10	5	15	1	15	1	1
12	6	15	1	15	1	0
14	7	105	7	105	7	0
16	8	105	4	93	3	0
18	9	945	33	885	28	0
20	10	945	15	885	15	0
22	11	10395	162	9405	140	0
24	12	10395	52	9405	41	0
26	13	135135	623	124215	555	0

allocator. Branching pairs the smallest unmatched index with each available partner (or with itself when the fixed slot is open and m is odd); this ordering beats largest-first by a factor of about fifteen in nodes. The engine was validated against an independent Python enumeration for all $m \leq 13$ (identical solution sets and counts).

Theorem 3 (computational). *The only fully symmetric no-three-in-line configurations with $n \leq 64$ occur at $n = 2, 4$, and 10 . Guy–Kelly Conjecture I holds through $n = 64$.*

Every run completed; Table 2 lists node counts. Growth is a factor of roughly 3.4 per unit of m within a parity class, with even m consistently cheaper (no fixed-point branch). Extrapolating this single-threaded engine gives about an hour at $n = 70$, tens of hours at $n = 80$, and an infeasible 10^7 hours at $n = 108$; Section 6 addresses the gap.

5 Why the first-moment heuristic fails

Guy and Kelly’s case for finiteness in the unrestricted problem is a first-moment computation: the expected number of solutions among random point sets tends to zero. The analogous estimate here would be $E(m) \approx N_{\text{inv}}(m) \cdot e^{-\lambda(m)}$, where N_{inv} counts admissible involutions and $\lambda(m)$ is the mean number of violated constraints for a uniform random involution, treated as Poisson. Monte Carlo sampling (2,000–3,000 configurations per size) gives: the mean number of collinear *triples* fits $T(m) \approx 4.7 m \ln m$ cleanly across $5 \leq m \leq 60$; violations arrive in D_4 -orbits of bad lines with an observed clump factor of about 7.9, so the natural Poisson variable is the number of violated line-orbits, whose mean $\lambda(m)$ we measured directly. Table 3 compares.

The estimate predicts $e^{4.3} \approx 73$ solutions at $n = 40$ and 3×10^5 at $n = 108$, against a verified count of zero through $n = 64$. Unlike the unrestricted problem, where the first moment argues *for* the conjecture, here it argues *against* it and is simply wrong — by a factor growing without bound.

The explanation is visible once one conditions on Lemma 1. We enumerated *all* involutions satisfying the Sidon condition through $m = 20$ and recorded, for each, the exact number of

Table 2: Search-tree sizes for the exhaustive runs (ascending branch order); all runs complete with zero solutions for $m \geq 6$ except m omitted below 13 for space.

m	n	nodes	time	m	n	nodes	time
13	26	508	<0.01 s	23	46	130,870	0.95 s
14	28	415	<0.01 s	24	48	95,898	0.70 s
15	30	1,472	0.01 s	25	50	420,125	3.4 s
16	32	1,117	<0.01 s	26	52	316,188	2.6 s
17	34	4,396	0.02 s	27	54	1,388,368	13.2 s
18	36	3,281	0.02 s	28	56	996,843	10.1 s
19	38	12,871	0.07 s	29	58	4,604,519	48.9 s
20	40	10,283	0.07 s	30	60	3,353,107	37.9 s
21	42	40,626	0.27 s	31	62	15,551,991	197 s
22	44	30,765	0.22 s	32	64	11,348,059	136 s

Table 3: The first-moment estimate versus reality. $\ln N_{\text{inv}}$ is exact; λ is the Monte Carlo mean number of violated line-orbits; the naive prediction is $\ln E = \ln N_{\text{inv}} - \lambda$.

m	n	$\ln N_{\text{inv}}$	λ	$\ln E_{\text{naive}}$	solutions
6	12	2.71	2.18	+0.5	0
8	16	4.65	3.42	+1.2	0
10	20	6.85	5.23	+1.6	0
14	28	11.81	9.12	+2.7	0
20	40	20.30	16.01	+4.3	0
32	64	39.80	32.32	+7.5	0
54	108	81.05	68.37	+12.7	0 (reported)

violated lines (Table 4). The class minimum — call it the *defect floor* $D(m)$ — is not merely positive: it climbs a staircase $8 \rightarrow 16 \rightarrow 24 \rightarrow 32$, stepping up roughly every five or six values of m , with odd m (one fixed point) consistently one step below neighboring even m . The survival probability is not $e^{-\lambda}$; on the visible range it is exactly zero, because the constraint system has a hard extremal core that random-model reasoning cannot see. This is the reduced-model form of the near-miss drift Flammenkamp observed in the raw grid data, and it suggests where a proof should come from.

Conjecture 2 (defect floor). *Let $D(m)$ be the minimum number of violated lines over all involutions of $\{1, \dots, m\}$ satisfying Lemma 1. Then $D(m) \geq 8$ for all $m \geq 6$, and $D(m) \rightarrow \infty$.*

Either statement implies Conjecture I for the corresponding range; the first is verified for $6 \leq m \leq 20$ by the enumeration above and implicitly for $m \leq 32$ by Theorem 3. Because the violated families realizing the minima are lines of slope ± 1 , central lines, and other small-slope families — all described by linear conditions on $(i, \sigma(i))$ — Conjecture 2 is a statement in additive combinatorics about label-distinct involutions, plausibly approachable by discrepancy or Fourier-analytic methods applied to the permutation matrix of σ .

Table 4: Exact enumeration of all Lemma-S survivors: counts, mean violated lines, and the class minimum (defect floor).

m	n	survivors	mean violated lines	$D(m)$
5	10	1	0.0	0
6	12	1	16.0	16
7	14	7	13.7	8
8	16	4	25.0	16
9	18	33	27.8	8
10	20	15	35.7	16
11	22	162	40.4	16
12	24	52	46.1	16
13	26	623	55.1	16
14	28	257	61.2	24
15	30	3,628	70.1	16
16	32	1,589	78.5	24
17	34	23,334	85.2	16
18	36	11,417	91.1	24
19	38	172,853	100.6	24
20	40	75,375	108.5	32

6 Beyond $n = 108$

6.1 Certified SAT

The involution model yields a compact propositional encoding. Variables are candidate orbits: pairs $\{i, j\}$ with $i < j$, plus singletons when m is odd — 1,540 variables at $n = 110$. Constraints: exactly one chosen orbit contains each index; a binary clause $(\neg a \vee \neg b)$ for every pair of orbits whose combined point set contains a collinear triple (this subsumes Lemmas 1 and 2, all 2+1 and 2+2 collisions, and the at-most-one-fixed rule; precomputation is a trivial $O(10^6)$ pair check at $m = 55$); and a ternary clause for each line receiving one point from each of three orbits, generated line-major over lines of the doubled-odd grid with at least three candidate points and pruned of clauses subsumed by binary conflicts. We estimate 10^6 – 10^7 clauses at $n \approx 110$, comfortably inside the range of CaDiCaL or Kissat with DRAT/LRAT proof logging, giving a machine-checkable nonexistence certificate for each n . Heule’s June 2026 SAT results on the much larger `rot4` and `rct4` classes [4] indicate the technology is more than adequate; the full class is the smallest of the symmetric classes. Implementation checks: $n = 10$ must be satisfiable with the unique model $\sigma = (1\ 5)(3\ 4)$, fixed point 2; every other even $n \leq 64$ must be unsatisfiable, cross-checked against Theorem 3. A cardinality relaxation of the same encoding (minimize the number of violated lines) extends the $D(m)$ table of Conjecture 2 far past $m = 20$.

6.2 Sidon-first enumeration

The Lemma-S survivor counts 1, 1, 1, 0, 1, 1, 7, 4, 33, 15, 162, 52, 623, 257, 3628, 1589, 23334, 11417, 172853, 75375 for $m = 1, \dots, 20$ grow by a factor of roughly five to seven per $\Delta m = 2$, vastly below $(m - 1)!!$. A search that enumerates label-distinct involutions as its outer loop — label bookkeeping is a single bitmask — and applies geometric checks lazily prunes the dominant constraint at near-

zero cost and should improve on the present engine by orders of magnitude. The engine here already supports splitting at the root for parallel or time-sliced execution, and a CUDA port has direct precedent in Riley’s enumeration of `rot4` classes [4].

7 Conclusions

The full-symmetry class reduces exactly to involutions with a Sidon-type label condition carrying most of the constraint weight; the known solution list is complete through $n = 64$ by independent verification; and the data locate the true obstruction in a growing extremal defect floor rather than in probabilistic thinning — indeed the standard heuristic points the wrong way here, which we regard as the most interesting single finding. The two live paths are a certified-UNSAT campaign, which should pass $n = 108$ with modest hardware, and a proof of the defect-floor conjecture, which would settle Guy–Kelly Conjecture I outright.

References

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